DATA COMMUNICATIONS AND PARALLEL PROCESSING C3200

End of term 1 test 1 Time: 1 hour 30 minutes

Answer any four questions.

QUESTION ONE

a) A wave has a frequency of 60 Hz and a length of 24 cm. Determine how fast it is moving (3 marks)

b) Given that the bit time is 10 ns, calculate the data rate? (3 marks)

c) You are given two codes, NRZ and RZ. Which one will transfer more bits per second? Explain your answer giving an example (4 marks)

d) Optic fibre s made up of three major components:

(i) light source

(ii) transmission

(ii) detector

Explain the function performed by each of the three (3 marks)

e) Three general frequencies used in wireless transmission are:

(i) 3 kHz - 1 GHz

(ii) 1 GHz - 300 GHz

(iii) $3 \times 10^{11} - 2 \times 10^{14}$

Explain the application area of each (3 marks)

QUESTION TWO

a) A wave signal has a frequency of 4 GHz and is traveling over a distance of 5 km. Calculate the anticipated signal loss due to attenuation? (3 marks)

b) A radio wave is travelling to GEO Satellite and back at a speed of 3×10^{12} meters per second. Calculate the time it will take to reach the receiver station (DOWN LINK) (2 marks)

c) Data Communications emphasizes on speed and may seem that anyone designing a data communication system would choose parallel transmission rather than serial.

d) Give two reasons why most data communication systems still use the serial mode. (2 marks)

e) On transmission order Ethernet technology specifies that data be sent byte big-endian and bit little-endian. Explain the meaning of this and use three bytes to illustrate your explanation. (3 marks)

f) Asynchronous transmission has a lot of disadvantages compared to synchronous transmission. Give reasons why it is still widely used despite these disadvantages, citing concrete and reasonable examples. (2 marks)

g) Asynchronous Communication technology is standardized by EIA. Give at least four EIA specifications for RS-232 asynchronous transmission (4 marks)

e) You are given the information below for asynchronous and synchronous transmission overheads:

Asynchronous transmission overheads:

- 1 start bit
- 1 stop bit
- 1 parity bit

Synchronous transmission overheads:

- Two SYN bytes
- two bytes CRC

(i) At first glance, it would appear that synchronous transmission has more transmission over heads. Explain with the help of diagrams why this is not true. (2 marks)

(ii) Use a 10 K byte data to compare the transmission efficiency of asynchronous and synchronous transmission. (2 marks)